Free Stock Data for Python Using Yahoo Finance API

In this post, I'm going to explore the use of integrating with the Yahoo Finance API via Python code.

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Mar 21, 2020·3 min read

Photo by Rick Tap on Unsplash

What is Yahoo Finance API?

Yahoo Finance is a media platform that provides financial news, data about stock quotes, press releases, and financial reports. And all the data provided by Yahoo Finance is free. Yahoo Finance API is the API that Yahoo provides to fetch financial information.

Yahoo deprecated their Finance API in 2017. So you can see many websites talking about alternatives for Yahoo Finance API. However, the python library **yfinance** offers a temporary fix to the problem by scraping the data from Yahoo! Finance and returning the data in the DataFrame format. So you can still use Yahoo Finance to get free stock market data.

Yahoo Finance provides access to more than five years of daily OHLC price data. Also, you can get minutes OHLC data for recent days.

Get Start

The Yahoo Finance API provides access to the information about:

- finance summaries like earnings, balance sheet.
- stocks historical prices.
- stock actions (including splits and dividends).

All this information is from the Yahoo Finance official website.

```
# To install yfinance before you use it.
> pip install yfinance
```

Rate Limitation

There're some limitations by making the call to Yahoo Finance API:

- Using the Public API (without authentication), you are limited to 2,000 requests per hour per IP (or up to a total of 48,000 requests a day).

I'm not sure it's precisely for Financial data. But please use time.sleep(1) to avoid your IP getting blocked.

Historical prices and chart

You have to call `Ticker` module to access ticker data. Here is an easy example to retreive five days of `MSFT` stock OHLC data:

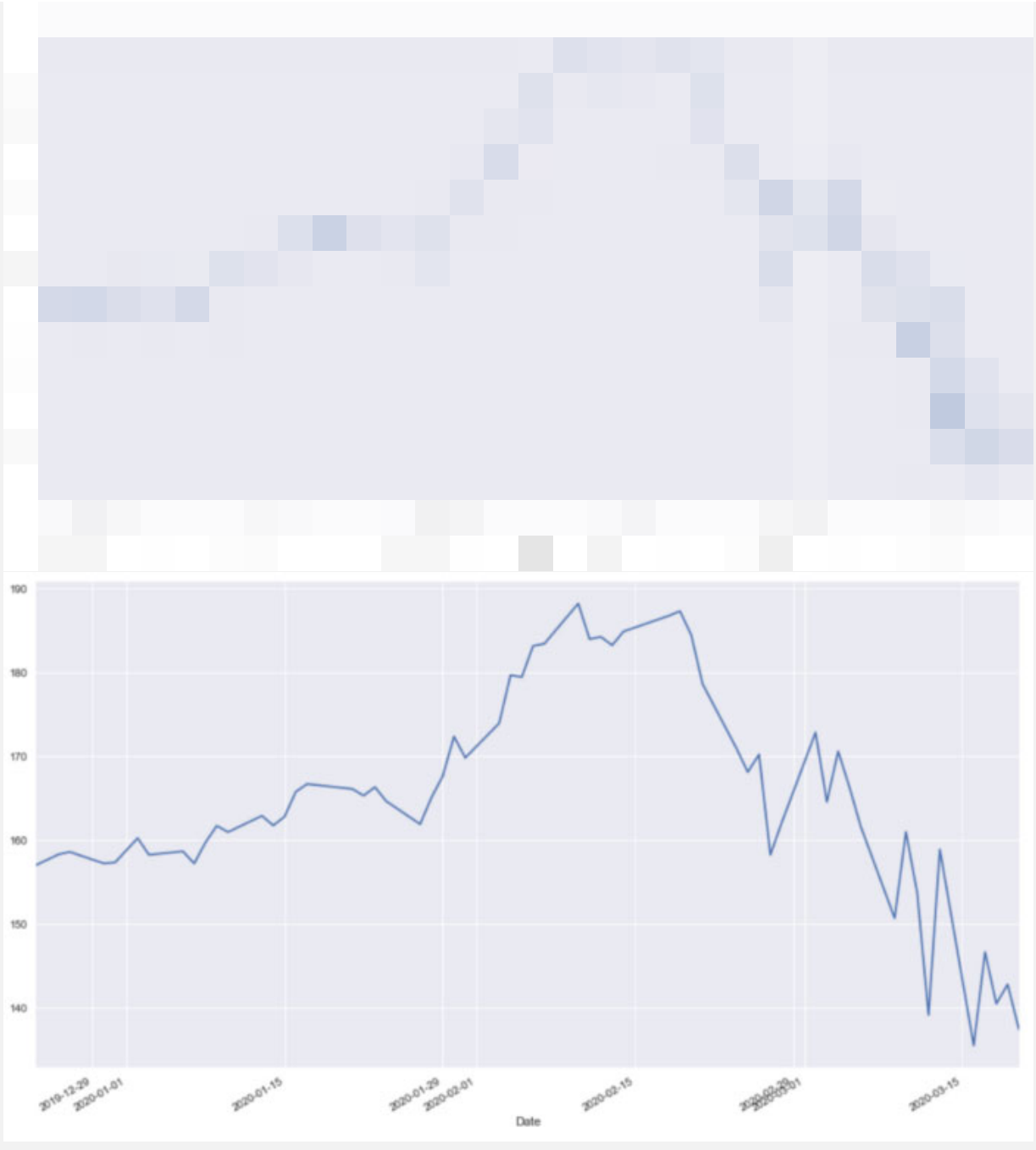
Now you can get the result like below:

There are more params you can set for history() method:

Plot historical prices

If you haven't installed yet, you can run commands to install matplotlib and seaborn.

```
pip install matplotlib
pip install seaborn
```



Export data as CSV

Also, you can export data as a CSV for analyzing.

This function is setting **auto_adjust** defaults to True. You can change the value if you don't want it.

Financial fundamentals

You can quickly check some financial data like dividends and splits. You need to change the ticker length of history to get financial fundamentals data.

Updated 2020/03/21: The earnings and balance sheet information is not available now. May need another API to find out the data.

In Conclusion

In this article, we had a look at how simple to get stock market data from Yahoo Finance. You can start write analysis behind the data. If you have any questions, please ask them below.